

SD5008
Research and Analysis for Design

WEEK 03

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Schedule

Week	Date & Place	Topic	Milestones
1	03/09/2026 TU201	**[Lecture]** Introduction to Design Research **[Special Lecture]** Prof. Jaden Park **[Tutorial]**	- Group arrangement after lecture
2	10/09/2026 V322	**[Library Workshop]** **[Lecture]** What & Why: Identifying and defining research dimensions; Research Through Design **[Tutorial]**	- Initial project idea / topic - Research purpose - Begin background & precedent review
3	17/09/2026 TU201	**[Lecture]** How #1: Approaches of Qualitative Research & Quantitative Research **[Tutorial]**	- Continue background review - Draft research question / creative proposition
4	24/09/2026 TU201	**[Lecture]** How #2: Data gathering mechanisms + AI **[Tutorial]**	-Submit **Research study project presentation 1** + Research topic + Literature mapping + Inquisitive direction + Research modalities in consideration
-	01/10/2026	***No Class***	- Initiate data gathering process



In today’s class:

1. Lecture “How: Approaches of Qualitative Research & Quantitative Research”

2. Specialism Studio Meeting with tutors

How: Research Methodologies

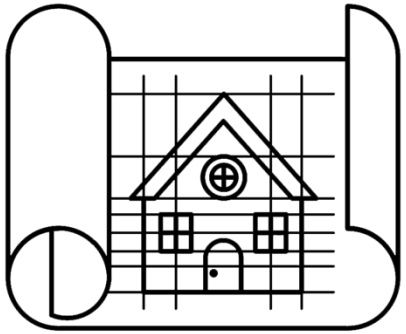
METHOD

- A way of acting or performing tasks, particularly in a systematic or routine way—an organized approach toward achieving a goal.



METHODOLOGY

- A body of methods employed in a particular activity
- A structured, predefined approach for executing a communication design project



Types of Research – Info Sources

PRIMARY RESEARCH

- Direct & Hands-On
- Gathering first-hand evidence to answer specific, project-bound questions and uncover fresh user insights.
- Collecting brand-new, original data directly from the source.

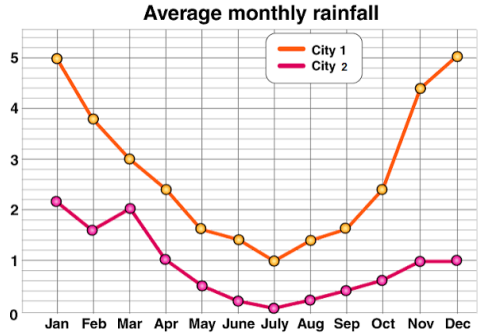
SECONDARY RESEARCH

- Context & Strategy
- Building essential background knowledge, evaluating existing visual strategies in similar contexts, and grounding in established, field-tested evidence.
- Analyzing, interpreting, and summarizing research and data that others have already gathered.

TERTIARY RESEARCH

- Overviews & Indexing
- Organizing and synthesizing primary and secondary sources to provide a high-level summary.
- Getting a quick overview of a topic, mapping out broad cultural or project contexts, and locating specific primary and secondary sources to investigate more deeply.

Types of Research - Data

Qualitative Research	Gathers words, opinions, and experiences through interviews or chats to understand thoughts and reasons.	It is appropriate for gathering deep personal feelings, opinions, and experiences instead of statistics. 
Quantitative Research	Gathers numbers and measurements using surveys, tests, or math to track trends and test ideas.	It relies entirely on hard numbers, percentages, and math to find patterns. 



QUALITATIVE APPROACH

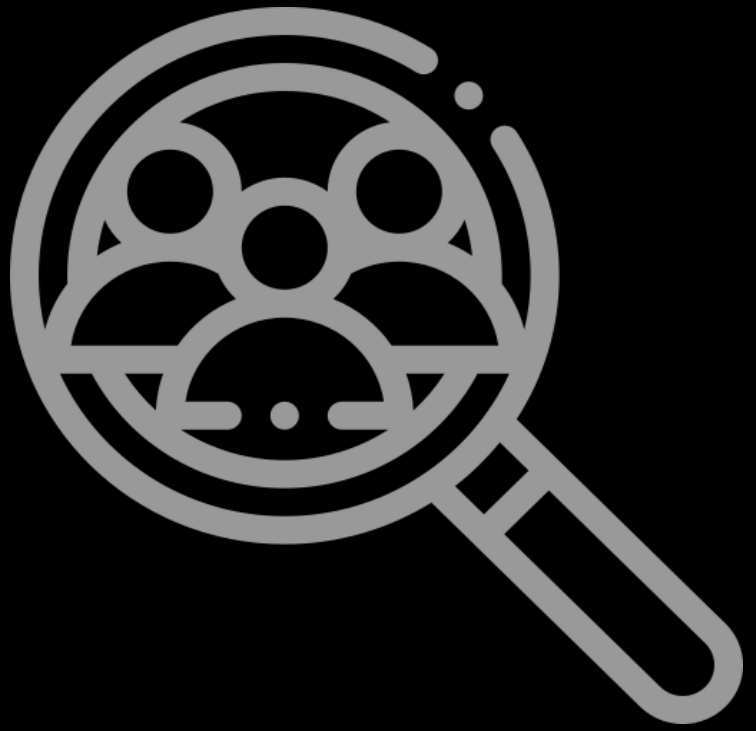
Qualitative Approach

- Qualitative approach engages with a problem that can be best understood by **exploring a phenomenon**.
- Qualitative approach values uniqueness of individuals and individual situations.
- It collects data in a natural setting sensitive to the people and places under study.
- It develops patterns / themes from data.
- Human experiences are multi-faceted, context-dependent, and impossible to reduce to a single objective measurement. Each exploration touches a different part of a complex systemic problem, offering a localized truth that contributes to a complete understanding.



Observations

- Observation is a qualitative method.
- It is not merely “looking” at a space or a group of people.
- It is a structured, intentional, and empathetic process of grasping visual and behavioral information.
- As we need to bridge the gap between real experience and creative creation, observation serves as the ultimate reality check: it reveals what people *actually do* versus what they say they do or what designers assume they do.



Observation

[A] Passive Observation (Fly-on-the-Wall)

- Observing users in their natural environment without interacting, speaking, or interfering with their activities.

Strengths:

- Captures authentic, unprompted human behavior.
- Reduces the risk of users changing their behavior because they are being evaluated.
- Very useful for identifying baseline patterns, physical bottlenecks, and spatial navigation habits.



Observation

[B] Participant Observation

- Actively joining the activity or environment alongside the participants, experiencing the context firsthand while simultaneously observing others.

Strengths:

- Fosters deep empathy by placing the designer directly ‘in the user's shoes.’
- Allows immediate, contextual questioning (asking “Why did you choose that?”) while the activity is taking place.
- Captures sensory and emotional nuances that passive observation misses.



Observation

[Z] Field Ethnography

- Immersive observation in the community *over time*, involving field notes, sketching, unstructured interviews, and artifact collection.

Strengths:

- Uncovers deep cultural values, social structures, and systemic communication breakdowns.
- Generates rich, highly context-aware data that prevents superficial design solutions.



Observation

[Y] Digital Ethnography

- Observing human behavior, language, visual culture, and communication in digital spaces, social media, communities, and forums.

Strengths:

- Uncovers deep cultural values, social structures, and systemic communication breakdowns.
- Generates rich, highly context-aware data that prevents superficial design solutions.



Observation

[X] Visual Ethnography

- Relying on photography, video recording, or visual mapping to capture environments, visual artifacts, and contextual behaviors.

Strengths:

- Provides rich visual evidence that is ideal for creating mood boards, persona journey maps, and compelling stakeholder presentations.
- Helps bridge cultural or language barriers through visual storytelling.



Observation

[W] Autoethnography

- A researcher uses their own personal, lived experience (*auto*) as primary empirical data to analyze and understand.

Strengths:

- Accesses implicit, non-verbal, and emotional dimensions of an experience through multimodal data.
- Enables research within private, highly specialized, or sensitive environments where third-party observation would be disruptive or ethically complex.
- Ideal for exploratory pilot phases or practice-led studies.



Observation

Observation Type	Passive Observation	Participant Observation	Field Ethnography	Digital Ethnography	Visual Ethnography	Autoethnography
Strengths	• Captures authentic, unprompted user behavior • Minimizes participant behavior changes • Ideal for mapping spatial, navigation, and baseline physical habits	• Builds deep empathy by experiencing context firsthand • Enables real-time, contextual questioning during tasks • Captures immediate sensory, physical, and emotional nuances	• Uncovers deep cultural values and systemic communication issues • Delivers rich, context-aware qualitative data	• Easy access to global, remote, or niche online communities • Yields searchable, archived visual and textual data	• Provides rich assets for journey maps and stakeholder presentations • Easily overcomes language and cultural barriers	• Enables to grasp deep tacit & emotional insight • Enables research within private, highly specialized, or sensitive environments
Weaknesses	• Cannot ask participants “why” an action was taken • High risk of misinterpreting user intent or motivation	• Active participation distracts from taking detailed field notes • Observer effect can alter normal participant behavior	• Extremely time- and resource-intensive • Produces vast unstructured data that is slow to synthesize	• Lacks physical, environmental, and non-verbal cues • Online behavior is frequently performative or curated	• Static photos miss temporal flow, movement, and sequence • Self-documented photo probes can be staged by participants	• Lack of Generalizability • Demands critical self-scrutiny with systematic data coding and validation

Interviews

- Interview is a qualitative method.
- It is a guided conversation designed to uncover how people construct, interpret, and navigate information.
- Where observation reveals what people do, interviewing reveals *why* they do it, providing designers with deeper context.



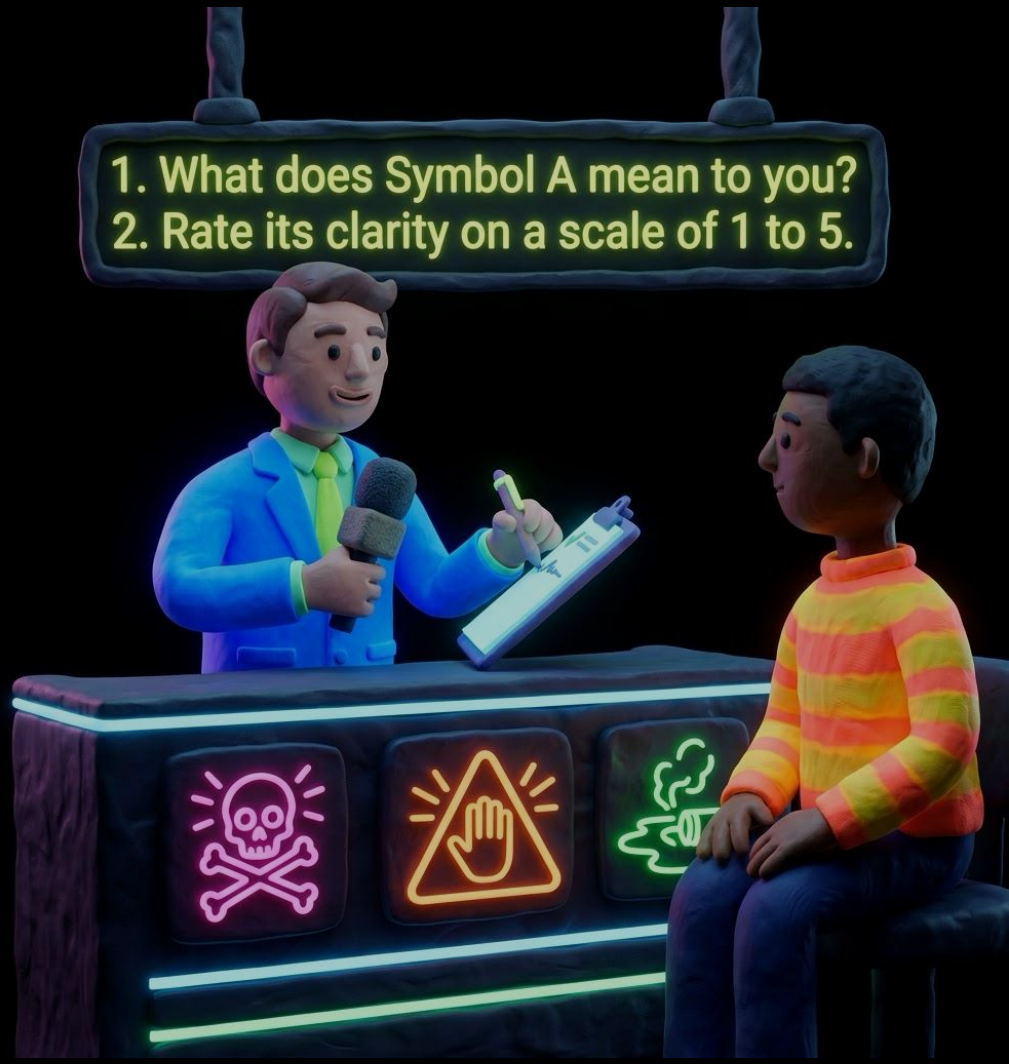
Interview

[I] Structured Interview

- A standardized interview involving asking a fixed set of pre-scripted questions in a strict order

Strengths:

- High consistency makes data easy to code, compare, and analyze across large participant groups.
- Reduces researcher bias and ensures every topic is addressed uniformly.



Interview

[II] Semi-Structured Interview

- An interview guided by a pre-set list of questions, while allowing the conversation to flow naturally and adapt based on participant responses

Strengths:

- Balances structural focus with creative flexibility, making it the most common interview type in design research.
- Enables researchers to pursue surprising tangents and uncover hidden user frustrations or visual literacy gaps.



Interview

[III] Unstructured Interview

- An open, conversational inquiry centered on a broad topic or theme without a pre-set list of questions

Strengths:

- Helps build high rapport and trust, allowing sensitive or complex cultural perspectives to emerge naturally.
- Ideal for early exploratory design research when the problem space is still undefined.



Interview

[z] Narrative Interview

- A type of interview that encourages participants to tell a continuous story about a specific event or journey.

Strengths:

- Captures rich emotional arcs, chronological context, and personal values embedded in everyday experiences.
- Reveals how users perceive communication touchpoints across an extended system or timeline.



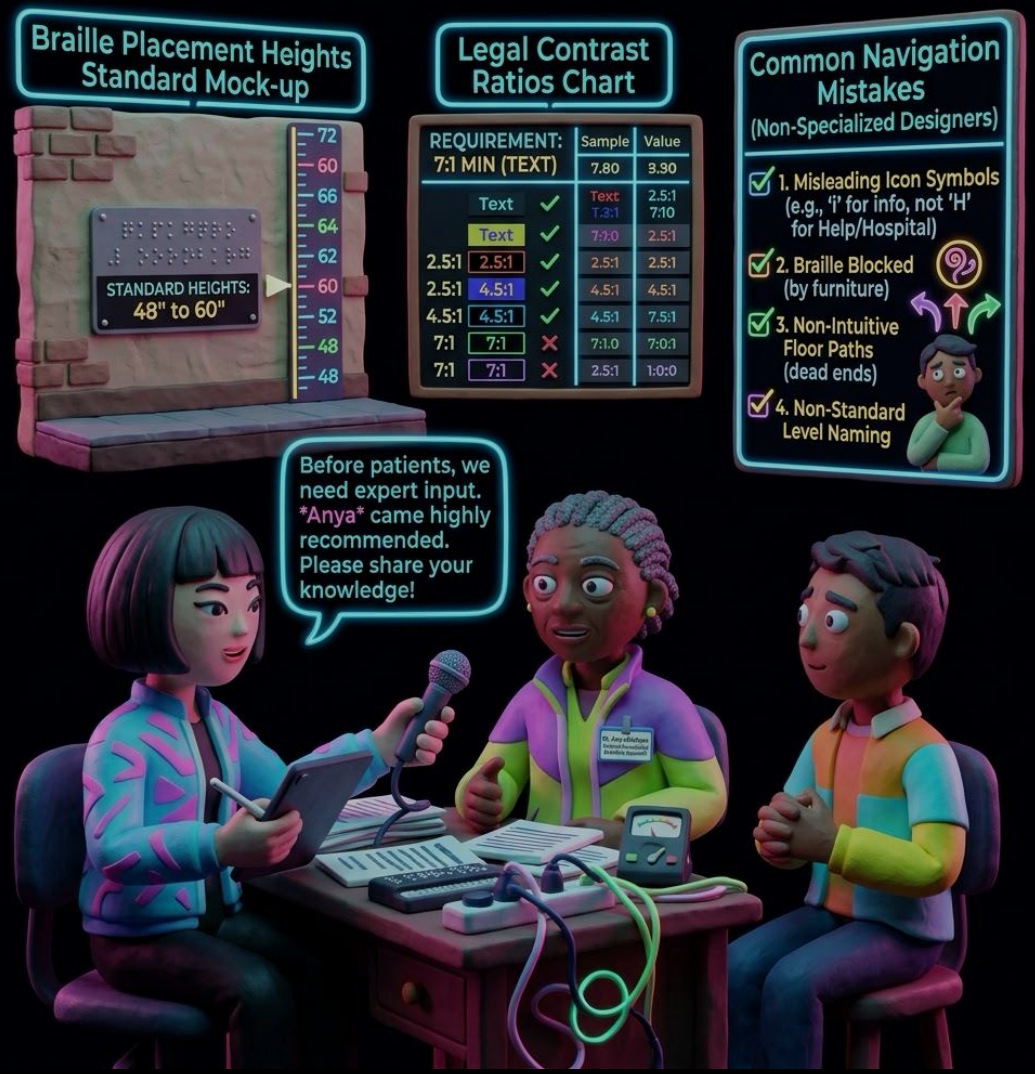
Interview

[y] Key Informant Interview (Expert Interview)

- A type of interview conducted with individuals who hold specialized knowledge, domain expertise, or institutional authority

Strengths:

- Quickly yields high-level systemic insights, historical context, technical constraints, and specialized vocabulary.
- Helps designers map out institutional rules and broader stakeholder ecosystems.



Interview

Interview Types	Structured Interview	Semi-Structured Interview	Unstructured Interview	Narrative Interview	Key Informant Interview
Strengths	• Highly consistent and repeatable • Easy to compare across large samples • Minimizes interviewer bias	• Balances structure with spontaneous probing • Generates rich, comparable qualitative data • Ensures core research questions are answered	• Uncovers unexpected organic insights • Builds strong trust and rapport early • Offers maximum conversational flexibility	• Reveals end-to-end user journeys • Pinpoints exact chronological pain points • Captures deep emotional arcs and motivations	• Quickly provides high-level domain expertise • Explains complex regulatory/systemic contexts • Efficiently grounds initial project research
Weaknesses	• Inflexible; cannot explore spontaneous insights • Superficial; fails to capture deeper “why” • Can feel transactional to participants	• Risk of asking leading follow-up questions • Time-consuming to transcribe and analyze • Requires careful guide preparation	• Difficult to compare across participants • High risk of going off-topic • Requires high facilitation skill	• Generates massive, lengthy transcripts • Participants may add irrelevant tangents • Time-intensive to code and synthesize	• Risk of expert or institutional bias • Experts may misunderstand actual user realities • Cannot replace direct end-user feedback

Focus Group

- Focus group is a qualitative method.
- It works with a small, **targeted** group of participants (typically 6 to 8 people) and has them engage in a moderated discussion around a specific concept, service touchpoint, or visual prototype.
- Unlike individual interviews, focus groups leverage social interaction, allowing participants to build on each other's ideas, debate trade-offs, and expose shared cultural assumptions.



Focus Group

Core Purposes

- **Evaluating Concepts & Artifacts:** Testing low-fidelity prototypes, visual brand directions, or service blueprints to gauge immediate emotional and social reactions.
- **Uncovering Collective Norms:** Observing how participants negotiate meaning, align on values, or disagree about everyday habits and system friction.
- **Group Synergy & Co-Creation:** Participants build upon and challenge each other's ideas, surfacing shared perspectives and unexpected insights that rarely emerge in isolated 1-on-1 interviews.



Focus Group

Weaknesses

- **Susceptibility to Groupthink:** Vocal or high-status participants can easily dominate the discussion, pushing quieter members to conform rather than voice dissenting opinions.
- **Attitudinal, Not Behavioral:** Captures what people claim or believe they do in a social context; it is completely unreliable for measuring real-world interaction speed, usability, or spatial navigation.
- **Limited Individual Depth:** Provides significantly less time per participant compared to interviews, making it poor for exploring private, sensitive, or highly complex workflows.
- **Heavy Moderation Dependency:** Unbalanced facilitation, poorly phrased prompts, or subtle moderator bias can easily distort the resulting data.

Participatory Co-Design Workshop

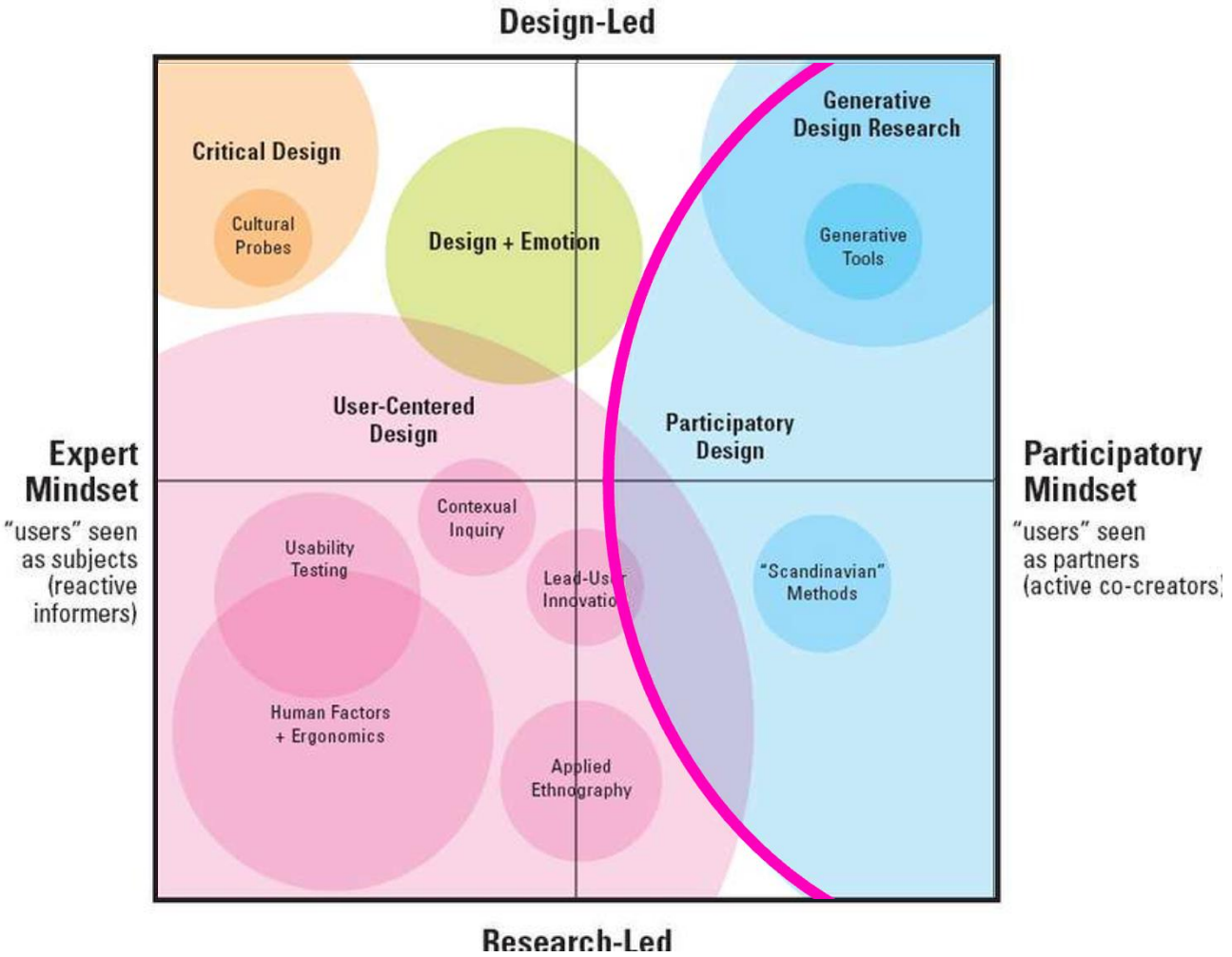
- Participatory co-design workshops treat end-users and stakeholders as active co-creators rather than passive research subjects; the *designer's role shifts from sole author to facilitator of collective creativity*.
- This **generative method** uses structured toolkits and physical activities to uncover unspoken human needs and co-formulate solutions.
- Uses tangible, non-intimidating creation tools, such as visual collaging, spatial mapping, card sorting, role-playing, and low-fidelity physical modeling, to explore what people feel and dream, accessing tacit knowledge and implicit emotional drivers.



Design Research Landscape

Participatory Design

- Attempts to engage the target users in the design process to ensure the final product or service meets their needs.
- Focuses on involving the people who are and/or will become the 'users' throughout the design development process.
- Physical artifacts are used as thinking tools throughout the process.



Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Participatory Co-Design Workshop

Phased Workshop Structure

- **Sensitizing & Warm-up:** Building psychological safety, breaking down hierarchy, and introducing the problem space.
- **Generative Co-Creation:** Working in small mixed groups using toolkits to build spatial, visual, or service concepts.
- **Reflection & Synthesis:** Presenting co-created artifacts back to the room to debate trade-offs, identify shared patterns, and evaluate feasibility.



Participatory Co-Design Workshop

Core Purposes

- **High Contextual Accuracy:** Co-created concepts directly reflect authentic lived experiences, significantly reducing the risk of deploying unusable design interventions.
- **Community Ownership & Empowerment:** Fosters genuine stakeholder involvement, transforming participants into decision-makers who protect, maintain, and expand the project long after the design team packs up and leaves.
- **Bypassing Verbal Limitations:** Tangible and visual toolkits allow participants with varying literacy, language fluency, or technical backgrounds to communicate complex, non-linear ideas equitably.



Participatory Co-Design Workshop

Weaknesses

- **Resource-Intensive Preparation:** Demands significant time and material investment upfront to design customized toolkits, manage logistics, and recruit representative participant groups.
- **Facilitation & Dominance Bias:** Loud, high-status, or extroverted participants can easily hijack group activities, requiring highly trained facilitators to protect quieter voices and maintain balance.
- **Data Synthesis Complexity:** Produces vast amounts of messy, unstructured qualitative data (sticky notes, audio recordings, physical models) that take substantial time to code, organize, and validate.
- **Idealism vs. Technical Constraints:** Participant-generated ideas often prioritize unconstrained feature wishlists that may conflict with real-world engineering, financial, or institutional limitations.



QUANTITATIVE APPROACH

Quantitative Approach

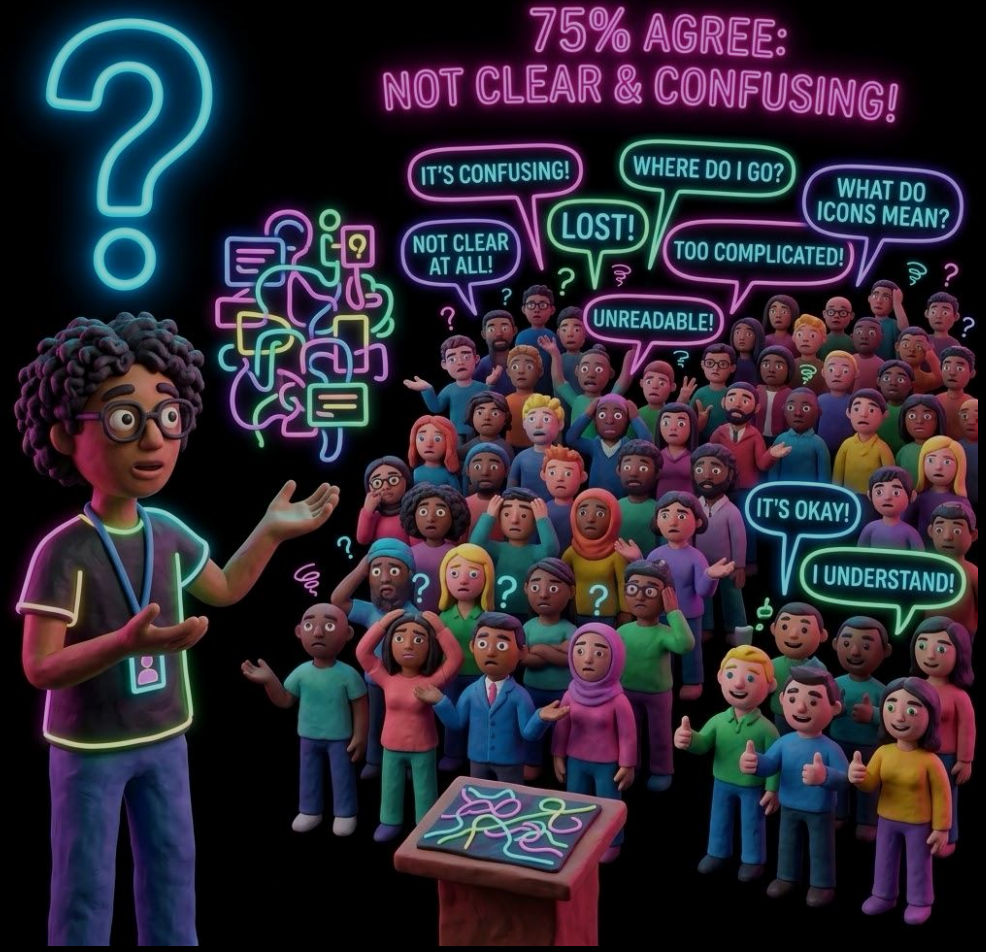
- Quantitative approach focuses on collecting, counting, and statistically analyzing numerical data.
- It is used to identify broad patterns, test specific hypotheses, and answer questions like “how many,” “how much,” or “how strongly related.”
- It provides *objective evidence* on how a large group of users interacts with, understands, or evaluates a design system.
- Its **aim is to generalize** findings to large populations.



Essentials for Quantitative Approach

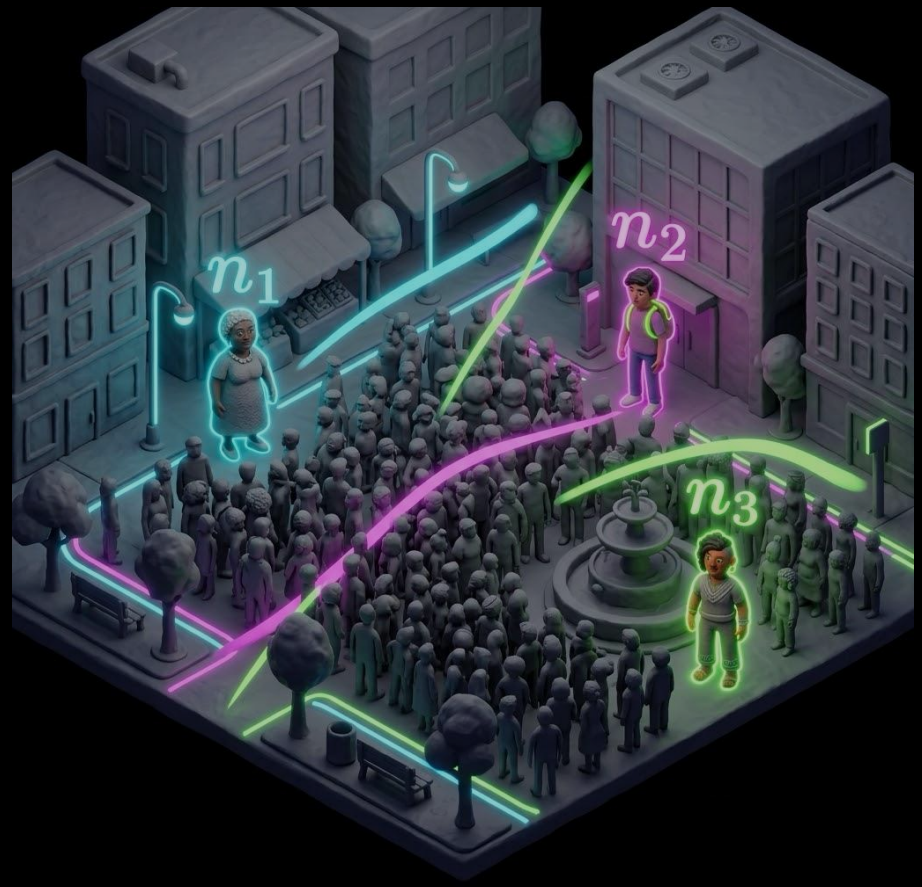
- **Measurable Questions:** Questions must focus on quantifiable relationships rather than open-ended feelings. (e.g., “75% of the participants found the signage to be not clear and confusing”).

- **Translating Design into Data:** Instead of asking if a poster is “readable,” you *measure how many seconds it takes a person to read it*. Instead of guessing if people “trust” a brand logo, you ask them to *rate their trust on a 1-to-7 scale*.



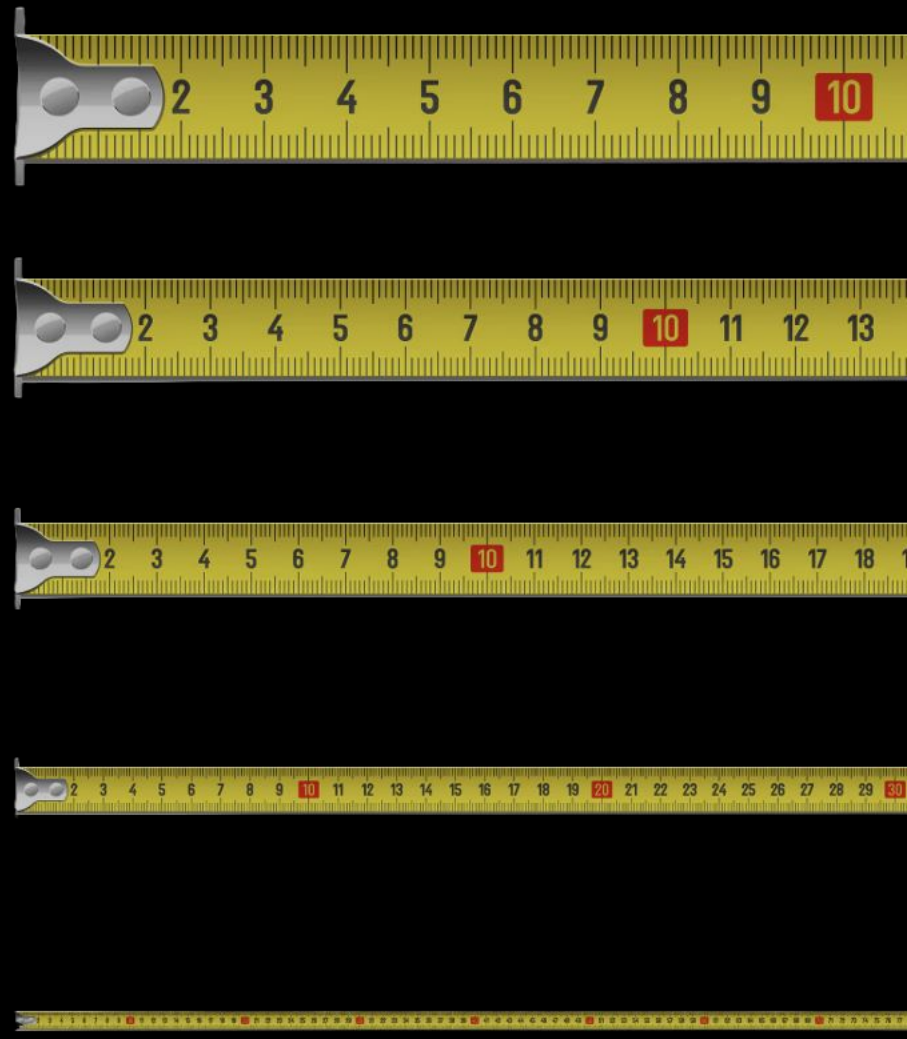
Essentials for Quantitative Approach

- **Sample Size & Target Audience:** In quantitative research, n stands for your total number of participants. To make your data believable, you need both volume and the right group:
  - n **= 3** → A coincidence (Not enough data to prove anything)
  - n **= 30** → The baseline (The minimum number to show a legitimate trend)
  - n **= 100+** → The big picture (Enough data to claim a pattern across a large population)
  - Matching the Target: Your participants must match your real-world user group, not just whoever was sitting next to you in class.



Essentials for Quantitative Approach

- **Reliability and Validity:**
  - **Reliability:** The measurement tool produces consistent results every time it is administered under identical conditions.
    - If your method gives random results every time, it isn't reliable.
  - **Validity:** The metric actually measures what it claims to measure.
    - Counting how fast someone clicks a button measures speed, not whether they actually liked the design.



Essentials for Quantitative Approach

- **Instrument Standardization:** Data collection tools (structured surveys, metric-based usability tests, analytics tracking) must remain identical for every participant to eliminate external factors and ensure fair comparison.
- **Bias Control:** Write survey questions and task prompts without emotional adjectives or leading hints so participants give their honest responses.



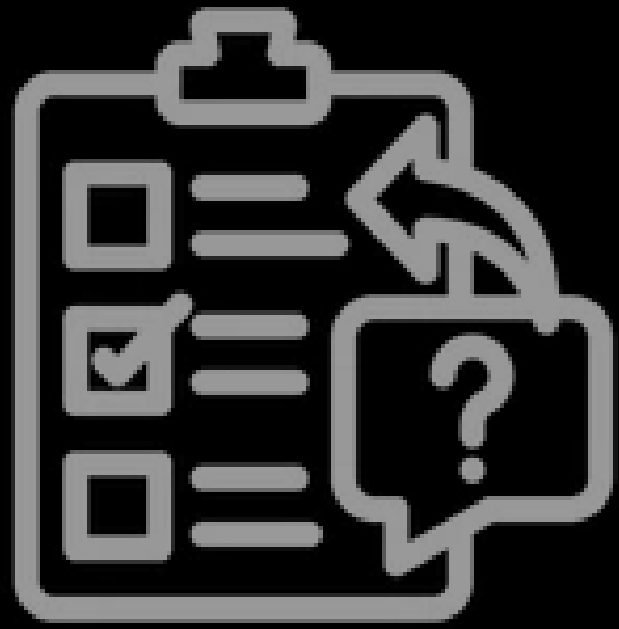
Essentials for Quantitative Approach

- **Data Processing Plan:** *Before gathering data, define what the responses will help you learn and how you will calculate and visualize them.*
- Think about whether to use basic descriptive statistics (e.g., percentages) or inferential statistics (e.g., correlations, variance testing) to derive conclusions.



Questionnaire [Survey]

- A questionnaire is a **hybrid research tool** - it can be qualitative & quantitative.
- It can gather *both descriptive human perspectives and numerical measurements* depending on how questions are structured.

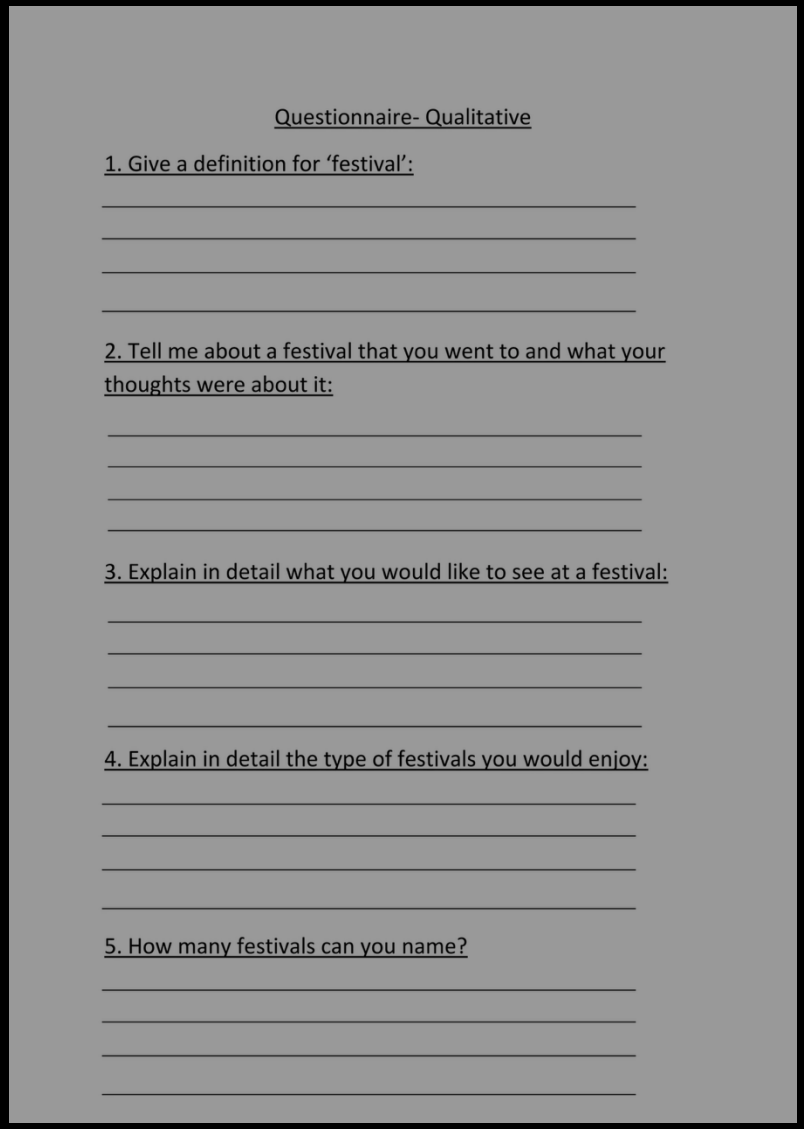


Questionnaire

Qualitative

– Exploring *Why* and *How*

- Uses open-ended text boxes that prompt participants to describe.
- It *resembles a structured interview*, but the key difference is that it is self-reported rather than conducted as a conversation
- Participants can describe feelings, explain behaviors, or reflect on experiences in their own words.
- This captures rich emotional context, personal terminology, and unexpected insights without forcing people into fixed categories.



Questionnaire- Qualitative

1. Give a definition for 'festival':

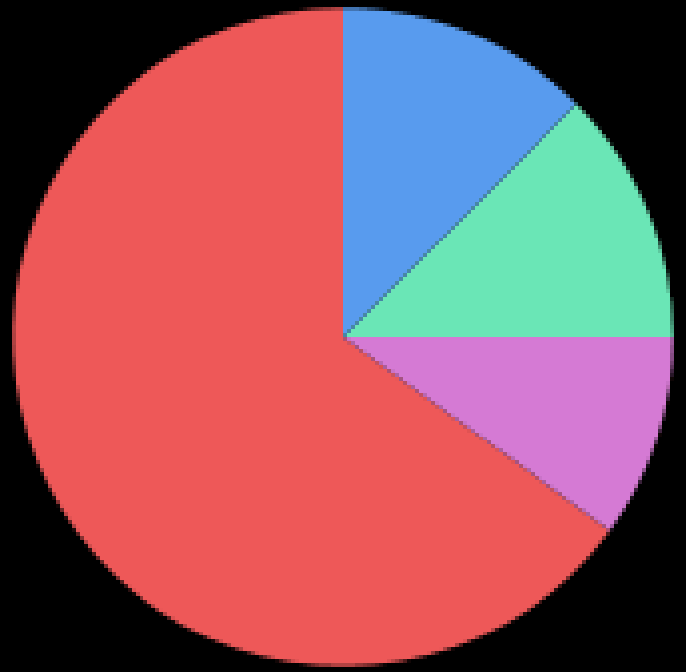
2. Tell me about a festival that you went to and what your thoughts were about it:

3. Explain in detail what you would like to see at a festival:

4. Explain in detail the type of festivals you would enjoy:

5. How many festivals can you name?

Questionnaire

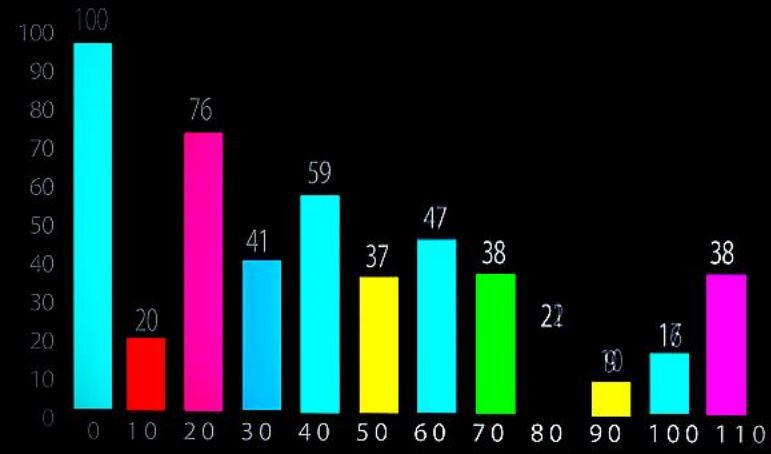


Quantitative

– Measuring *How Many* or *How Much*

- Uses closed-ended questions
  - ❑ Yes / No
  - ❑ Likert scales
  - ❑ Multiple-choice
  - ❑ Ranking choices



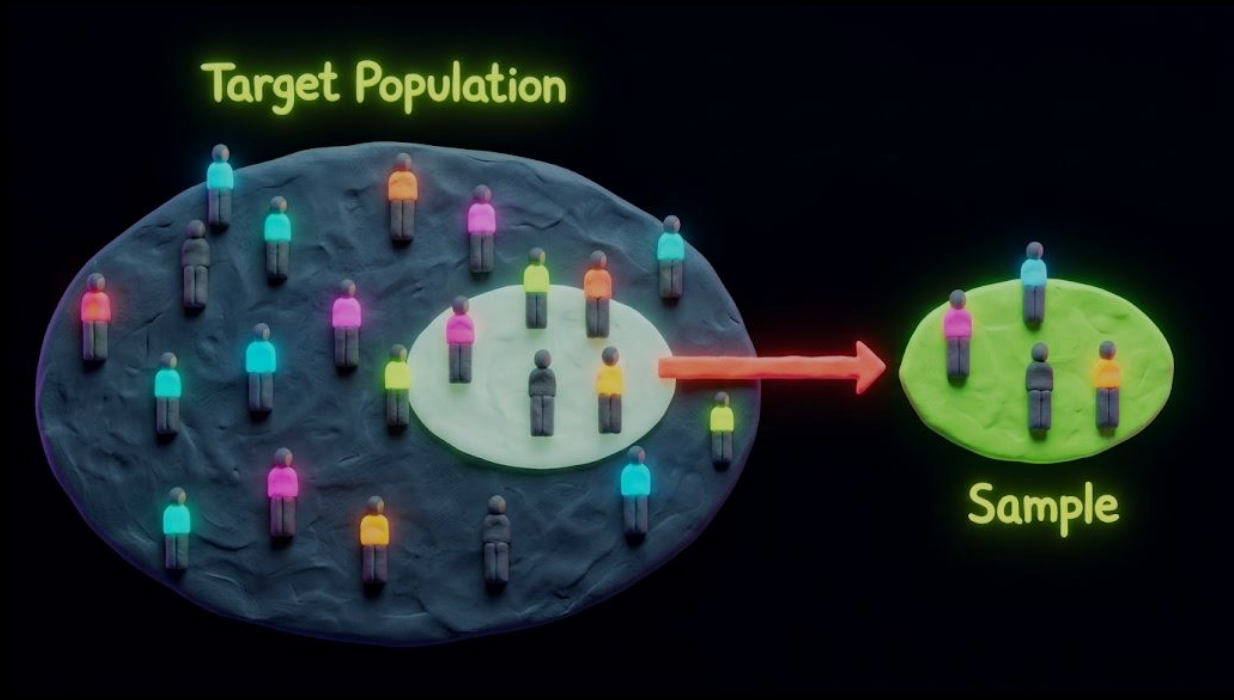


- Measures broad patterns, such as the percentage.

Questionnaire – Quantitative

Sampling

- A sample is a manageable group of individuals chosen from the target population to represent that entire group.
- Sample size (n) directly determines if a statistical baseline accurately represents population traits or if it's just random noise. Without sufficient sample size, the results lack statistical stability.
- Defining your sample determines if survey findings reflect a real pattern or just a few opinions.
- We must systematically construct the sampling framework before distributing any questionnaire.



Questionnaire – Sampling

Consideration #1

Target Population Boundaries

- **Define Specific Parameters:** Move beyond broad categories. Instead of targeting "*public transit users*," specify "*daily subway commuters in Hong Kong aged 65 and older*," based on the research question.
- **Set Exclusion Rules:** Filter out participants whose unique background would distort the data



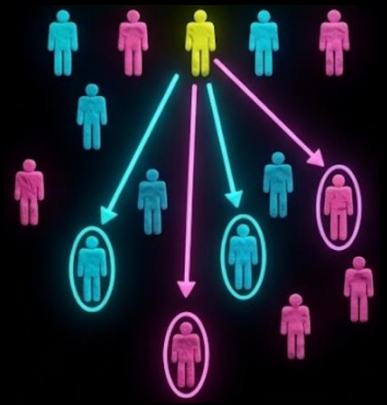
Questionnaire – Sampling

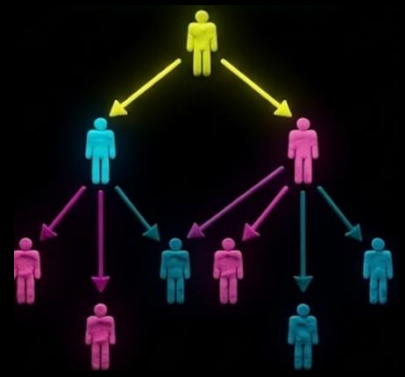
Consideration #2

Sampling Strategy

Convenience Sampling	Purposive Sampling	Snowball Sampling	Random Sampling
Recruiting easily accessible people (e.g., classmates, roomates). High risk of bias; best reserved for quick pilot testing.	Intentionally selecting participants who fit precise behavioral or context criteria (e.g., individuals who use screen readers daily).	Asking participants to forward the survey to peers in their network; ideal for reaching hard-to-access or niche communities.	Giving every person in a defined population an equal chance of selection; the gold standard for generalizability, though challenging for short-term student projects.







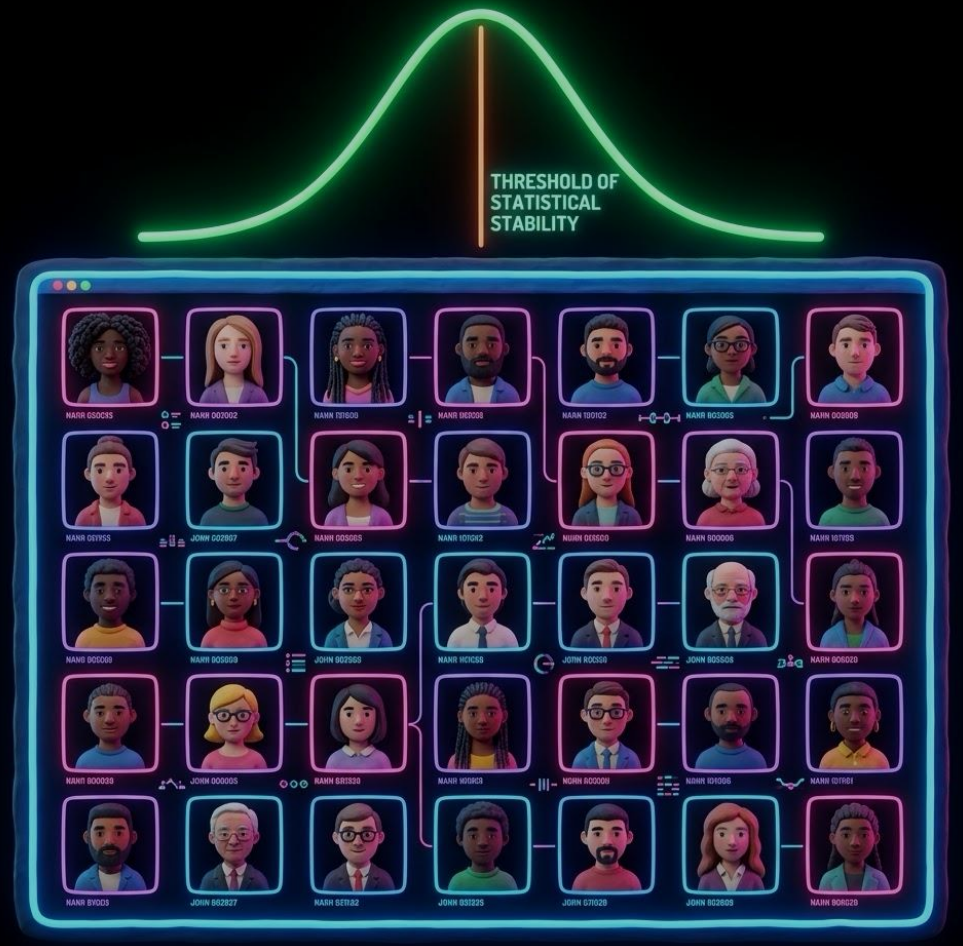


Questionnaire – Sampling

Consideration #3

Sample Size Requirements

- **Statistical Baseline ($n \geq 30$):** The absolute minimum sample size needed to demonstrate standard statistical distributions and reduce the impact of random outliers.
- **Broad Population Generalization ($n \geq 100+$):** Necessary if you intend to generalize and claim that your design findings apply to an entire campus, city, or user demographic.



Questionnaire – Sampling

Consideration #4

Demographic Balance & Representativeness

- **Proportional Ratios:** Mirror the real-world demographics of your target group across key variables such as age, language fluency, technical literacy, and gender.
- **Avoid the "Design Bubble":** Don't rely solely on fellow design students as your sample; their heightened design literacy will obscure usability issues that everyday users experience.



Questionnaire – Sampling

Consideration #5

Bias Identification & Control

- **Self-Selection Bias:** Acknowledge that people who voluntarily complete surveys often hold stronger positive or negative opinions than the silent majority.
- **Channel Bias:** Distributing a survey exclusively via a digital channel (e.g., Instagram or Discord) automatically excludes non-digital populations, distorting age and lifestyle metrics.

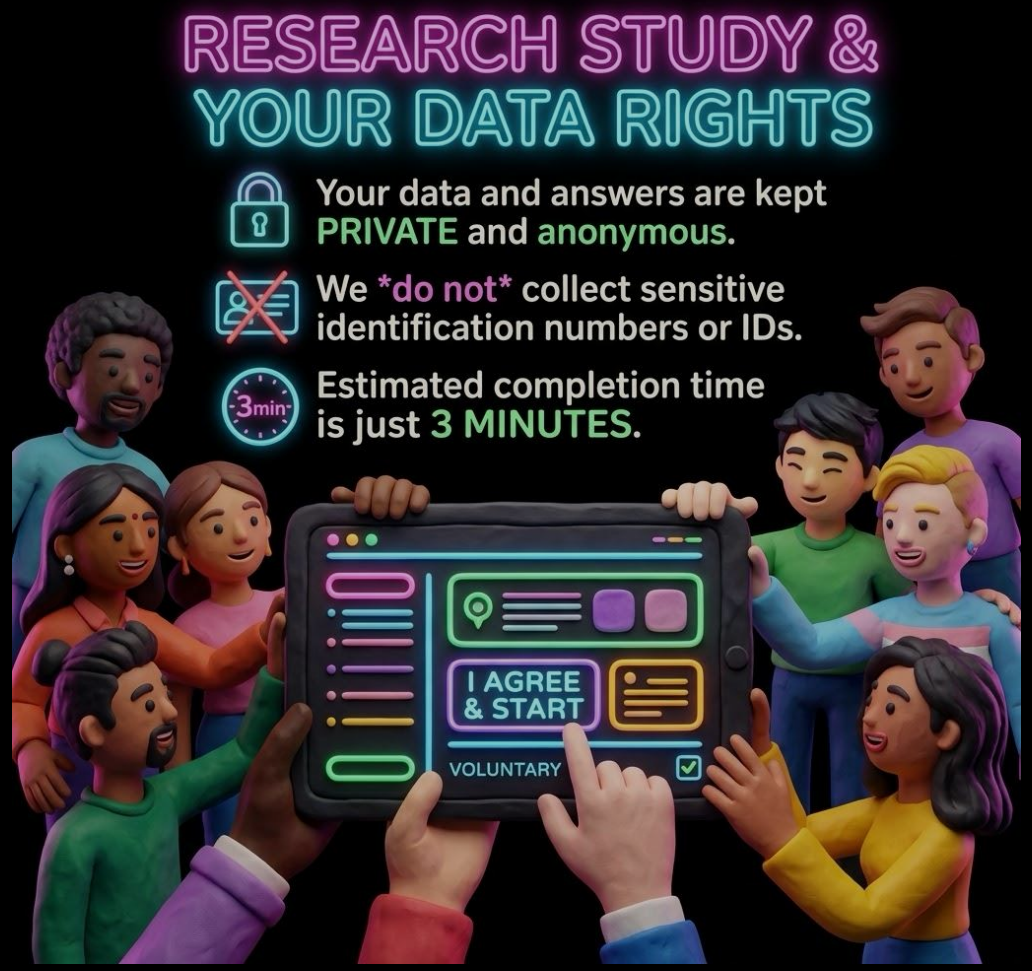


Questionnaire – Sampling

Consideration #6

Ethical Access & Voluntary Participation

- **Informed Consent:** Clearly state the study's purpose, estimated completion time, data handling, and confidentiality protections on the first page.
- **Equitable Accessibility:** Ensure digital forms are screen-reader accessible and formatted clearly for low-vision participants, or provide printed/multilingual alternatives where appropriate.

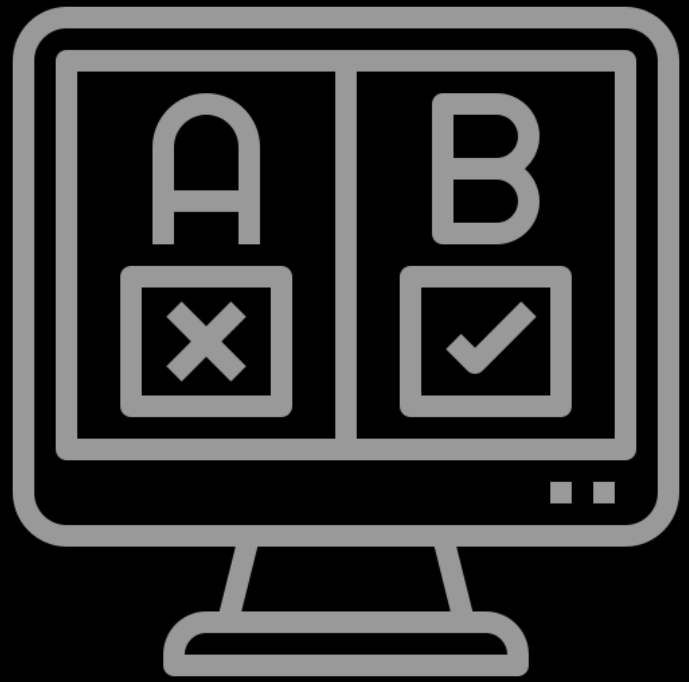


Questionnaire vs. Survey

Feature	Questionnaire (method)	Survey (methodology)
Definition	A structured document or form containing a set of questions (open-ended, closed-ended, or mixed).	An end-to-end research framework that uses questionnaires to collect, analyze, and synthesize data.
Scope	Narrow; focuses purely on question phrasing, scale selection, and visual layout.	Broad; includes population sampling, data collection strategy, statistical analysis, and theoretical synthesis.
Primary Goal	To gather raw answers from an individual participant.	To identify trends, generalize patterns, or measure opinions across a population.
Sample Size	Approximately 30	Minimum 100
Usage	To gather rich text, support contextual inquiries (exploratory), identify usability breakdowns, conduct pilot testing.	To identify general user preferences, basic ratings, or compare two design options (generalization through statistical validation).

A/B Testing

- A/B testing (or split testing) is a quantitative experimental method.
- Two versions of a single variable - Variant A (the control) and Variant B (the treatment) - are presented at random to comparable user groups to measure which option performs better against a defined metric.
- It isolates one specific element at a time (such as typographic contrast, button placement, or navigation hierarchy) while keeping all other design elements strictly identical.
- Relies on sample size (n) and probability math to prove that performance differences are driven by the design intervention rather than random chance.



A/B Testing

Core Purposes

- **Bypasses the "Say vs. Do" Gap:** Evaluates actual human behavior in live environments, eliminating the social desirability bias common in surveys and focus groups.
- **High Data Credibility:** Replaces subjective aesthetic debates in design critiques with empirical numbers, making a clear case for interaction choices.
- **Low Operational Risk:** Enables designers to run controlled live experiments on small user slices without risking full-scale product re-launches.
- **Precise Hypothesis Testing:** Provides unambiguous clarity when deciding between two specific visual or structural choices.



A/B Testing

Weaknesses

- **Absence of Qualitative Context:** Reveals what users did, but cannot explain why they did it or how they felt during the experience.
- **Detail Trap:** Focuses heavily on fine-tuning minor UI details (e.g., button colors), which can distract from addressing fundamental architectural or systemic design flaws.
- **High Sample Size Requirements:** Demands substantial user traffic (n) to achieve statistical significance, making it unviable for low-traffic platforms or niche public services.

Ethical Approval Before Research

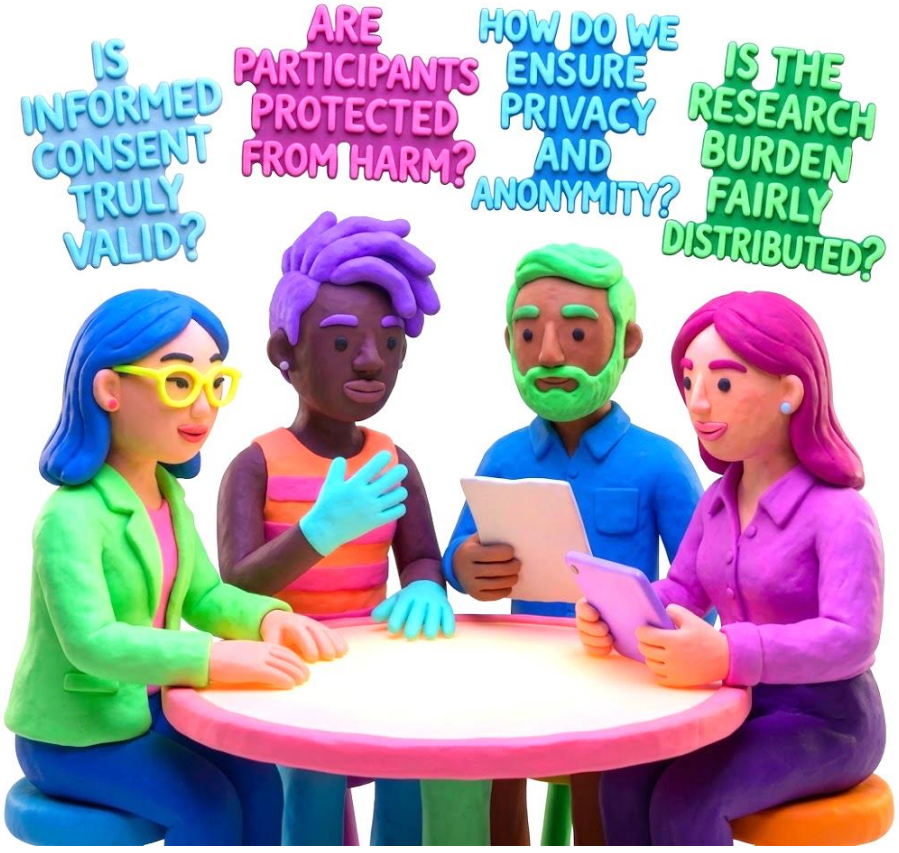
- Before start taking actions to conduct research, we should consider whether what we are about to do is ethical.

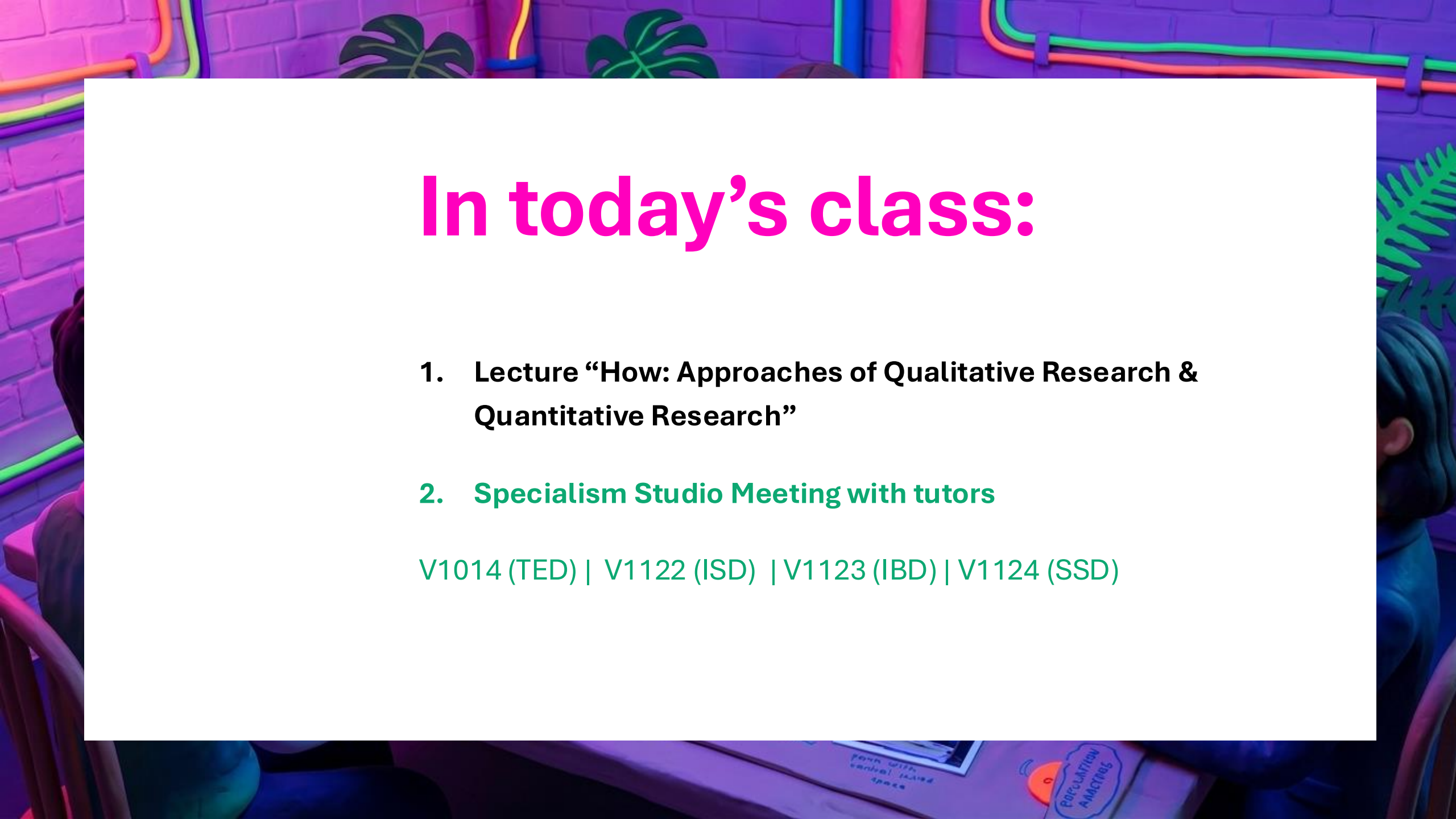
  ✓ The subject matter of the study

  ✓ The research methods and procedures

- Ethical approval must be secured before collecting any data to protect human participants from physical, psychological, or privacy-related harm. Without prior clearance, any gathered data is academically invalid and cannot be used in final design projects, presentations, or research publications.

- Institutions need to ensure our projects comply with university policies and legal data privacy regulations, protecting both you and the institution from legal liability.





In today’s class:

1. **Lecture “How: Approaches of Qualitative Research & Quantitative Research”**

2. **Specialism Studio Meeting with tutors**

V1014 (TED) | V1122 (ISD) | V1123 (IBD) | V1124 (SSD)